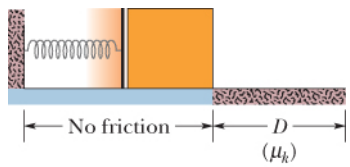
Homework #11

1) A child whose weight is 267 N slides down a 6.10 m long playground slide that makes an angle of 20° with the horizontal. The coefficient of kinetic friction between slide and child is 0.10. (a) How much energy is transferred to thermal energy? (b) If she starts at the top with a speed of 0.457 m/s, what is her speed at the bottom?

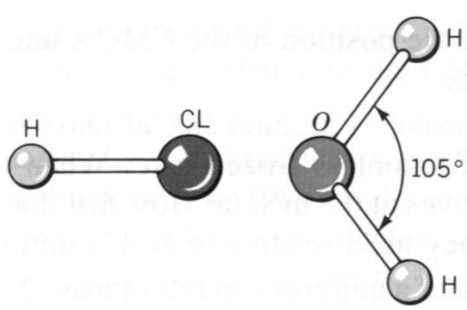
2) In the figure below, a 3.50 kg block is accelerated from rest by a compressed spring of spring constant 640 N/m. The block leaves the spring at the spring's relaxed length and then travels over a horizontal floor with a coefficient of kinetic friction μ_k = 0.25. The frictional force stops the block in distance D = 7.80 m. What are (a) the increase in the thermal energy of the block–floor system, (b) the maximum kinetic energy of the block, and (c) the original compression distance of the spring?



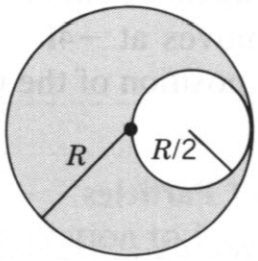
3) A 25 kg bear slides, from rest, 12.0 m down a pine tree, moving with a speed of 5.6 m/s just before hitting the ground. (a) What change occurs in the gravitational potential energy of the bear–Earth system during the slide? (b) What is the kinetic energy of the bear just before hitting the ground? (c) What is the average frictional force that acts on the sliding bear?

4) During a rockslide, a 520 kg rock slides from rest down a hillside that is 500 m long and 300 m high. The coefficient of kinetic friction between the rock and the hill surface is 0.25. (a) If the gravitational potential energy U of the rock–Earth system is zero at the bottom of the hill, what is the value of U just before the slide? (b) How much energy is transferred to thermal energy during the slide? (c) What is the kinetic energy of the rock as it reaches the bottom of the hill? (d) What is its speed then?

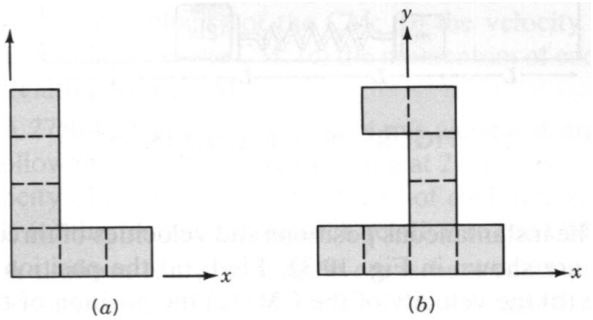
5) Locate the center of mass of the following molecules: (a) The HCl molecule has a dumbbell shape, as shown below, with the atoms separated by 1.30×10^{-10} m. The mass of the H atom is 1 u and that of the Cl atom is 35 u. (b) The H_2O molecule is also shown below. The mass of the O atom is 16 u. The H and O atoms are separated by 1×10^{-10} m and the bond angle is 105°.



6) A uniform disk of radius R has a circular hole of radius $R/2$ removed as shown in the figure. Locate the center of mass with respect to the center of the original disk. (Hint: Treat the hole as an object of negative mass.)



7) (a) The shape in (*a*) below is cut from a uniform sheet of material. Each square of side 2 cm has a mass of 10 g. Locate the center of mass with respect to the origin. (b) Repeat (a) for the shape in (*b*).



8) What is the average momentum of an avalanche that moves a 40-cm-thick layer of snow over an area of 100 m by 500 m over a distance of 1.00 km down a hill in 5.5 s? Assume a density of 350 kg/m3 for the snow.